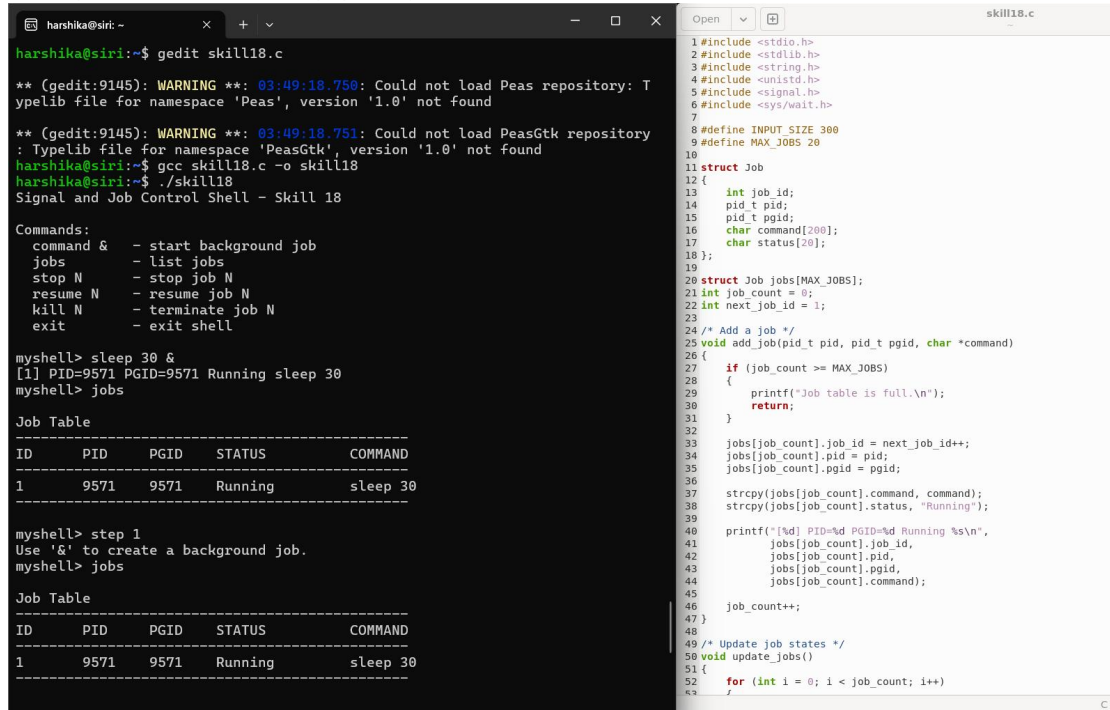


Skill - 18

1. Signals, Processing groups



```
harshika@siri: ~  
harshika@siri:~$ gedit skill18.c  
  
** (gedit:9145): WARNING **: 03:49:18.750: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:9145): WARNING **: 03:49:18.751: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found  
harshika@siri:~$ gcc skill18.c -o skill18  
harshika@siri:~$ ./skill18  
Signal and Job Control Shell - Skill 18  
  
Commands:  
command & - start background job  
jobs - list jobs  
stop N - stop job N  
resume N - resume job N  
kill N - terminate job N  
exit - exit shell  
  
myshell> sleep 30 &  
[1] PID=9571 PGID=9571 Running sleep 30  
myshell> jobs  
  
Job Table  
-----  
ID      PID      PGID     STATUS    COMMAND  
-----  
1       9571     9571     Running   sleep 30  
-----  
  
myshell> step 1  
Use '&' to create a background job.  
myshell> jobs  
  
Job Table  
-----  
ID      PID      PGID     STATUS    COMMAND  
-----  
1       9571     9571     Running   sleep 30  
-----
```

```
1#include <stdio.h>  
2#include <stdlib.h>  
3#include <string.h>  
4#include <unistd.h>  
5#include <signal.h>  
6#include <sys/wait.h>  
7  
8#define INPUT_SIZE 300  
9#define MAX_JOBS 20  
10  
11struct Job  
12{  
13    int job_id;  
14    pid_t pid;  
15    pid_t pgid;  
16    char command[200];  
17    char status[20];  
18};  
19  
20struct Job jobs[MAX_JOBS];  
21int job_count = 0;  
22int next_job_id = 1;  
23  
24/* Add a job */  
25void add_job(pid_t pid, pid_t pgid, char *command)  
26{  
27    if (job_count >= MAX_JOBS)  
28    {  
29        printf("Job table is full.\n");  
30        return;  
31    }  
32  
33    jobs[job_count].job_id = next_job_id++;  
34    jobs[job_count].pid = pid;  
35    jobs[job_count].pgid = pgid;  
36  
37    strcpy(jobs[job_count].command, command);  
38    strcpy(jobs[job_count].status, "Running");  
39  
40    printf("[%d] PID=%d PGID=%d Running %s\n",  
41        jobs[job_count].job_id,  
42        jobs[job_count].pid,  
43        jobs[job_count].pgid,  
44        jobs[job_count].command);  
45  
46    job_count++;  
47}  
48  
49/* Update job states */  
50void update_jobs()  
51{  
52    for (int i = 0; i < job_count; i++)  
53    }
```